

HOMEWORK 5

Exercise: 1, 6, 7, page 66;
Exercise: 4, 5, 10, 11, 12, 13, page 73-74;

In the following, we fix a field F .

Problem 1. Let $A \in \text{Mat}_{m \times n}(F)$ be a matrix of row rank 1.

- (1) Show that there is a matrix $u \in \text{Mat}_{m \times 1}(F)$ and a matrix $v \in \text{Mat}_{1 \times n}(F)$ such that $A = uv$.
(Here uv means the matrix product of u and v).
- (2) If $m = n$, show that $A^2 = \text{tr}(A)A$.

The next problem is a generalization of the above one.

Problem 2. Let $A \in \text{Mat}_{m \times n}(F)$ be a matrix of rank k . Show that there is a matrix $C \in \text{Mat}_{m \times k}(F)$, $R \in \text{Mat}_{k \times n}$ such that $A = CR$.

(Hint: use elementary row operations. Here one can take R to be the nonzero rows of the RRE matrix of A . What can you say about C ? For an integer p with $p < \text{rank}(A)$, is it possible to find matrices $C_1 \in \text{Mat}_{m \times p}(F)$ and $R_1 \in \text{Mat}_{p \times n}(F)$ such that $A = C_1 R_1$? We will go back to this question in future.)

Problem 3. Let F be a field of characteristic 0. We view elements of F^3 as column vectors. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & -2 \\ 0 & -1 & 2 \end{bmatrix}$$

and the linear operator $T : F^3 \rightarrow F^3$ given by $T(\alpha) = A\alpha$. Compute $\ker(T)$ and $\ker(T^2)$. Moreover, show that $\ker(T^k) = \ker(T)$ for any positive integer k . Here $T^2 = T \circ T$, and $T^k = T \circ T \circ \dots \circ T$ (k -times).

Problem 4. Let V be a vector space over F and let $T : V \rightarrow V$ be a linear map. Suppose $\ker(T) = \ker(T^2)$. Show that $\ker(T^k) = \ker(T)$ for any $k \geq 1$.

Problem 5. We have defined an equivalence relation on the set $\text{Mat}_{m \times n}(F)$ using row equivalence, namely, $A \sim B$ if A and B are row equivalent. Let $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$, the finite field with 3 elements. Consider the set $\text{Mat}_{2 \times 3}(\mathbb{F}_3)$. For each equivalence class in $\text{Mat}_{2 \times 3}(\mathbb{F}_3)$, find a representative of it. How many equivalence classes are there in $\text{Mat}_{2 \times 3}(\mathbb{F}_3)$?